SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.6: Area of a Part of a Circle — Teacher Reference (Pre-filled)

SUMMARY TABLE: CORE MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.6: Area of a Part of a Circle
Driving Question	DRIVING QUESTION: How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers, and artisans can make accurate plans and avoid waste?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon & Area of an Annulus**	Students observed a real-world anchoring phenomenon — a centre-pivot irrigation system in the Rift Valley, where water sweeps a large circular field but a concrete inner ring (around the pivot base) remains unwatered. The teacher displayed an aerial photograph and asked students to estimate what fraction of the total circular area actually receives water. Students noted that the watered region is a ring shape (annulus) between a large outer circle and a smaller inner circle.	The area of an annulus is calculated by subtracting the area of the inner circle from the area of the outer circle: $A = \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$, where R is the outer radius and r is the inner radius. This formula applies wherever a circular region has a circular hole — irrigation rings, round chapati pans with a central hole, or the flat face of a metal washer used by a Nairobi fundi.	The teacher should emphasise the difference-of-squares factorisation $\pi(R+r)(R-r)$ as an efficient mental-check form. Pedagogical note: cold-call at least three students to substitute different R and r values before assigning written practice. Stress that $R > r$ must always hold; a common error is reversing the subtraction and obtaining a negative area. Connect back to the driving question by asking: 'If the Rift Valley farmer needs to buy fertiliser only for the annular watered zone, how does this formula prevent waste?'
Lesson 2 **Area of a Sector: The Pivot's Sweep**	Students observed a short animation (or teacher-drawn diagram) showing the irrigation pivot arm rotating through various central angles — 90°, 120°, and 270° — and shading the swept 'pie-slice' region each time. They recorded how much of the full circle each coloured sector appeared to cover, making proportional estimates before any formula was introduced.	The area of a sector with central angle θ (in degrees) and radius r is $A = (\theta/360) \times \pi r^2$. The sector is a proportional fraction of the full circle, so the formula scales the full-circle area πr^2 by the fraction $\theta/360$. Students also encountered the radian form $A = \frac{1}{2} r^2\theta$ for reference. Kenyan context: a maize farmer in Nakuru whose pivot arm sweeps 270° can calculate exactly how many square metres of crop are irrigated.	The teacher should draw the analogy: 'Just as you cut a chapati or ugali into equal slices, the sector is one slice of the circle — the angle tells you how large a slice.' Pedagogical note: give students a 360° protractor cut-out and have them physically fold it to $\theta = 90°$ (quarter) and verify $A = \pi r^2/4$ before touching the formula. A frequent misconception is using $\theta/180$ instead of $\theta/360$; address this explicitly. Ensure students can rearrange to find r or θ when area is given.

Lesson 3 **Arc Length and Sector Perimeter: Fencing the School Garden in Nakuru**	Students observed a scaled plan of a school garden in Nakuru shaped as a sector (a quarter-circle plot). They were asked: 'The school wants to fence the curved edge and the two straight edges — how many metres of fencing wire are needed?' Students measured the two radii on the plan and estimated the curved boundary, noticing that the arc is shorter than the full circumference.	Arc length $l = (\theta/360) \times 2\pi r$. The full perimeter of a sector $= l + 2r = (\theta/360) \times 2\pi r + 2r$. Students distinguish arc length (a length, measured in metres) from sector area (measured in m^2). In radian form, $l = r\theta$. Kenyan artisan link: a jua-kali metalworker in Gikomba bending a curved steel strip to form the arc of a decorative gate panel uses exactly this calculation.	Pedagogical note: insist students write units at every step — a common error is quoting arc length in square units. Use the mnemonic 'arc length is a fraction of the full hoop' to reinforce $(\theta/360) \times 2\pi r$. Cold-call students to find the perimeter for $\theta = 60°$, $150°$, and $240°$ with $r = 7$ m (a value that cancels neatly with $\pi \approx 22/7$). Connect to the driving question: 'How does knowing exact arc length help a Nakuru fence contractor buy just enough wire and avoid waste?'
Lesson 4 **Area of an Annular Sector: Combining the Ring and the Wedge**	Students examined a diagram of the Laikipia irrigation system where the pivot arm does not cover the full 360° but only sweeps a wedge of angle θ, and the inner concrete base creates a hole — producing a wedge-shaped ring (annular sector). Students were asked to identify which earlier shapes this new region combines, drawing on Lessons 1 and 2.	The area of an annular sector $= (\theta/360) \times \pi(R^2 - r^2) = (\theta/360) \times \pi(R+r)(R-r)$. It is the area of the larger sector (radius R) minus the area of the smaller sector (radius r) at the same angle θ. Students recognise this as a composition of the annulus formula and the sector formula — a key modelling skill for the driving question's theme of combining sub-regions.	Pedagogical note: use a colour-coded diagram — shade the full outer sector in green, overlay the inner sector in red, and show that the annular sector is what remains. This visual is especially effective for learners who struggled with the abstract subtraction. Reinforce that both sectors must share the same central angle θ and the same centre. Ask: 'A Laikipia farmer's pivot arm has $R = 50$ m, $r = 5$ m, and sweeps 120° — what area is watered?' Work this together, then assign a parallel problem with Kenyan-contextualised values.
Lesson 5 **Area of a Segment of a Circle**	Students observed a chord drawn across a circle (representing a straight dam wall cutting across a circular reservoir near Lake Victoria). They identified two regions — the smaller segment (between the chord and the minor arc) and the larger segment — and were challenged to find the area of water stored in the smaller portion without measuring an irregular shape directly.	The area of a circular segment = (sector area) − (triangle area) $= (\theta/360)\pi r^2 - ½r^2\sin\theta$, which simplifies to $½r^2(\pi\theta/180 - \sin\theta)$. Students must correctly identify the central angle θ subtended by the chord, compute the sector, then subtract the isosceles triangle formed by the two radii and the chord. Kenyan context: engineers designing a small earth dam at a circular reservoir use this to estimate water volume per unit depth.	Pedagogical note: the most common error is forgetting to subtract the triangle, leaving students with only the sector area. Use a physical demonstration — cut a sector from card and fold along the chord to show the triangular flap that must be removed. Emphasise that $½r^2\sin\theta$ uses the sine of the full central angle, not half of it. Provide a WAIT TIME of at least 60 seconds after posing the Lake Victoria dam problem before cold-calling. Connect back: 'How does this formula help an engineer avoid over- or under-estimating the volume of water stored?'
Lesson 6 **Area of a Common Region Between Two Intersecting Circles**	Students observed two overlapping circular plots on a Kisumu urban farm plan — each plot irrigated by its own sprinkler — and identified the lens-shaped overlap where both sprinklers water the same ground. They were asked why a farmer would want	The common (lens-shaped) region between two intersecting circles of equal radius r, with centres distance d apart, equals the sum of two congruent circular segments: $A = 2 \times [½r^2(\theta - \sin\theta)] = r^2(\theta - \sin\theta)$, where θ is the central angle (in radians) subtended	Pedagogical note: this is the most conceptually demanding lesson. Begin by revisiting the segment formula from Lesson 5 and explicitly show that the lens is 'segment from circle 1' + 'segment from circle 2'. Use a Venn-diagram analogy —

	to calculate this overlapping area (to avoid double-counting when estimating water or fertiliser needs).	at each centre by the chord of intersection. For circles of unequal radii r_1 and r_2, each segment is computed separately using its own radius and central angle, then added. Students use the cosine rule to find θ from d, r_1, r_2.	students have seen overlapping sets; here the overlap has an area that can be measured. Cold-call at least four students at different steps: (1) finding the chord length, (2) finding θ using the cosine rule, (3) computing each segment, (4) summing. Stress accuracy in calculator use: ensure students work in consistent angle units (degrees or radians) throughout.
Lesson 7 **Mixed Application and Problem Solving: Combining Circle-Part Formulae**	Students were presented with a composite diagram representing a decorative metalwork gate panel designed by a Gikomba jua-kali artisan. The panel combined an annular sector, two segments, and a central full circle. Students first identified and labelled each sub-region before attempting any calculation — practising the decomposition strategy.	A composite circle-part figure can be decomposed into annuli, sectors, annular sectors, and segments. The total area is found by summing (or subtracting) the areas of constituent sub-regions using the correct formula for each. Learners select formulae strategically: they check whether a region requires addition or subtraction and verify that all radii and angles are correctly assigned to each sub-region.	Pedagogical note: this lesson functions as a structured review and synthesis session. Resist the urge to demonstrate all examples — instead, use a 'group jigsaw': assign each group one sub-region, have them compute its area, then assemble the total as a class. This mirrors real-world collaborative practice in Kenyan engineering offices. Common errors to anticipate: using the wrong radius for a sector within a larger composite, and failing to convert angles consistently. End by returning to the driving question and asking students to articulate — in their own words — how each formula learned across all lessons helps farmers, engineers, and artisans avoid waste.
Lesson 8 **Assessment and Consolidation: Applying Circle-Part Formulae to Real Kenyan Contexts**	Students reviewed the complete anchoring phenomenon — the Rift Valley centre-pivot irrigation system — now annotated with all sub-regions studied across the unit (annulus, sector, annular sector, segment, lens). They identified which formula governs each zone and discussed, as a class, how the full set of formulae together answers the driving question for farmers, engineers, and artisans across Kenya.	Students demonstrate consolidated mastery of all five circle-part area formulae: (1) Annulus: $\pi(R^2-r^2)$; (2) Sector: $(\theta/360)\pi r^2$; (3) Arc length and sector perimeter: $(\theta/360)\times 2\pi r + 2r$; (4) Annular sector: $(\theta/360)\pi(R^2-r^2)$; (5) Segment: $(\theta/360)\pi r^2-½r^2\sin\theta$; (6) Lens (common region): sum of two segments. They can select, apply, and combine these formulae to solve unseen Kenyan contextual problems.	Pedagogical note: the assessment task should be performance-based where possible — for example, presenting students with a scaled plan of a real (or realistic) Kenyan site (farm, school compound, artisan workshop floor) and asking them to calculate areas for a practical purpose such as material costing or irrigation planning. Provide a formula sheet for lower-attaining students to reduce cognitive load and focus assessment on application rather than recall. After marking, conduct a 10-minute whole-class error analysis targeting the two or three most common mistakes observed. Celebrate growth by asking students to compare their Lesson 1 predictions with their Lesson 8 solutions, linking back to the driving question as a closing reflection.